Unit 1 - Introduction to Database Systems

Database Management (DI03016041) - GTU Diploma IT Sem 3

16% weightage | 7 hours

How to use these notes: every topic is written in a descriptive, copy-ready way. **3 marks** = definition + 2-3 points. **4 marks** = add example or table. **7 marks** = write everything with diagram / example. Lines marked ★ show where the topic was asked in a GTU paper (Winter 2025 / Summer 2026). SQL examples use Oracle syntax.

GTU questions already asked from this unit (do these first)

- Asked in BOTH papers: W2025 Q1(a) [3M: Metadata, Schema, Database] + S2026 Q1(a) [3M: Data, Information, Metadata]
- Asked in S2026 Q1(c)OR [7M: advantages of DBMS over file system]
- Asked in W2025 Q1(b) [4M: disadvantages of DBMS]
- Asked in BOTH papers (7M each): W2025 Q1(c) [responsibilities of DBA] + S2026 Q1(c) [roles and responsibilities of DBA]
- BOTH papers: W2025 Q1(c)OR [7M: data abstraction + three-level ANSI-SPARC architecture] + S2026 Q1(b) [4M: data abstraction + 3 levels]

1.1.1-1.1.4 Data, information, database, DBMS, metadata, schema, instance

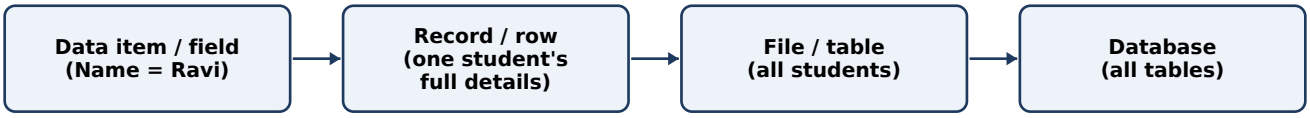
★ Asked in BOTH papers: W2025 Q1(a) [3M: Metadata, Schema, Database] + S2026 Q1(a) [3M: Data, Information, Metadata]

Term	Definition	Example
Data	Raw, unprocessed facts and figures that have no meaning by themselves	45, 'Ravi', 9876543210
Information	Processed, organised data that has meaning and is useful for decisions	'Ravi scored 45 out of 50 - Grade A'
Database	An organised collection of related data stored together, so that it can be easily accessed, managed and updated	College database: students, courses, marks
DBMS	Software that creates, stores, manages and controls access to the database	Oracle, MySQL, SQL Server, PostgreSQL
Metadata	**'Data about data'** - it describes the structure of the data: table names, column names, data types, sizes, constraints, relationships	Table STUDENT has columns Roll_no NUMBER(5), Name VARCHAR2(30)
Schema	The **overall logical design / structure** of the database (tables, columns, relationships, constraints). It is defined at design time and changes rarely	STUDENT(Roll_no, Name, City, Branch)
Instance	The **actual data stored in the database at a particular moment**. It changes every time data is inserted, updated or deleted	The rows currently in the STUDENT table

Easy way: **Schema = blueprint of a house; Instance = the people living in it today.** Data → (processing) → Information.

1.1.5-1.1.6 Data items, fields, records and data dictionary

Data is stored in a database in a hierarchy of units, from smallest to largest:



Term	Meaning	Example
Data item (field)	Smallest named unit of data; one column value	Roll_no, Name, City
Record	Collection of related fields about one entity (one row / tuple)	(101, Ravi, Surat, IT)
File (table)	Collection of records of the same type	STUDENT table
Data dictionary	A **catalogue (central repository) of metadata** maintained by the DBMS. It stores table and column names, data types, sizes, constraints, indexes, users and privileges, relationships. The DBMS uses it while executing every query; DBA uses it for management. In Oracle: USER_TABLES, USER_TAB_COLUMNS, USER_CONSTRAINTS.	Where the DBMS looks up that column 'City' of STUDENT is VARCHAR2(20)

1.1.3 Purpose of database system and applications

Before DBMS, organisations kept data in separate **files** (file-processing system). That caused many problems, which are the **purpose** of having a database system:

1. **Data redundancy and inconsistency** - same data stored in many files; when one copy is updated, others stay old.
2. **Difficulty in accessing data** - every new query needs a new program.
3. **Data isolation** - data scattered in files of different formats.
4. **Integrity problems** - rules (e.g. balance > 0) are buried in program code and hard to change.

5. **Atomicity problems** - a failure in the middle of a transfer leaves the data half-updated.
6. **Concurrent-access anomalies** - two users updating the same data at the same time give wrong results.
7. **Security problems** - hard to give each user access to only a part of the data.

Applications of database systems: banking (accounts, transactions), universities (students, results), airlines (reservations), e-commerce (products, orders), telecom (call records, bills), hospitals, library, railways, social media.

1.2 File-oriented system versus database system (advantages of DBMS)

★ Asked in S2026 Q1(c)OR [7M: advantages of DBMS over file system]

File system stores data in separate files managed by the operating system and application programs. **DBMS** stores data in an integrated database controlled by one software. Advantages of DBMS over a file system:

Point	File-oriented system	Database system (DBMS)	Advantage of DBMS
Redundancy	Same data repeated in many files	Data stored once and shared	Less duplication, saves space
Consistency	Copies get out of date	One copy, updated once	Data is consistent
Data sharing	Difficult; files tied to programs	Many users and applications share one database	Easy sharing
Data independence	Program depends on file format	Program is independent of storage structure	Change storage without changing programs
Integrity	Rules coded in every program	Constraints (PK, FK, CHECK) enforced centrally	Accurate, valid data
Security	Only OS-level, weak	Users, roles, privileges (GRANT / REVOKE)	Controlled access
Concurrency	Not controlled	Locking and transaction management	Safe multi-user access
Backup and recovery	Manual	Automatic backup, log-based recovery	Data safe after a crash
Querying	New program for each query	SQL - powerful query language	Fast, easy retrieval

Exam tip: For 7 marks write an introduction of both, then at least 6 advantages with a line each (or the full table).

Disadvantages of DBMS

★ Asked in W2025 Q1(b) [4M: disadvantages of DBMS]

1. **High cost** - DBMS software (e.g. Oracle) and powerful hardware are expensive.
2. **Complexity** - designing, setting up and managing a database needs skill and training.
3. **Need for trained staff** - a DBA and programmers must be hired, increasing salary cost.
4. **Size** - DBMS needs a lot of disk space and memory to run.
5. **Performance overhead** - for small, simple applications it can be slower than a plain file system because of the extra features.
6. **Single point of failure** - if the central database or DBMS fails, all applications stop; regular backup is a must.
7. **Security risk** - all data is in one place, so a successful attack can expose everything.
8. **Conversion cost** - moving from an old file system to a DBMS takes time and money.

1.3 Database Administrator: roles and responsibilities

★ Asked in BOTH papers (7M each): W2025 Q1(c) [responsibilities of DBA] + S2026 Q1(c) [roles and responsibilities of DBA]

The **Database Administrator (DBA)** is the person (or team) with **central control over the database and the DBMS**. The DBA makes sure the database is available, secure and running efficiently.

Roles and responsibilities of the DBA

Responsibility	What the DBA does
1. Schema definition	Creates the original database schema using DDL (tables, columns, keys)
2. Storage structure and access method	Decides how data is physically stored, creates indexes for fast access
3. Schema and physical-organisation modification	Changes the schema or storage when needs change (ALTER, reorganise files)
4. Granting authorisation	Decides who can access which data - creates users, roles, GRANT / REVOKE privileges
5. Security and integrity	Protects from unauthorised access; defines integrity constraints so data stays valid
6. Backup and recovery	Takes regular backups; restores data after a crash or disk failure; manages transaction log
7. Performance monitoring and tuning	Watches response time, tunes queries and indexes, manages disk space
8. Routine maintenance	Checks free disk space, installs upgrades / patches, archives old data
9. Communication with users	Talks to users and application programmers, understands requirements, gives training and support
10. Concurrency and data dictionary	Ensures many users can work together safely; maintains the data dictionary

Exam tip: A quick 7-mark answer: intro line + any 7 responsibilities with one line each. Add 'DBA has special privileges (e.g. SYS / SYSTEM in Oracle)'.

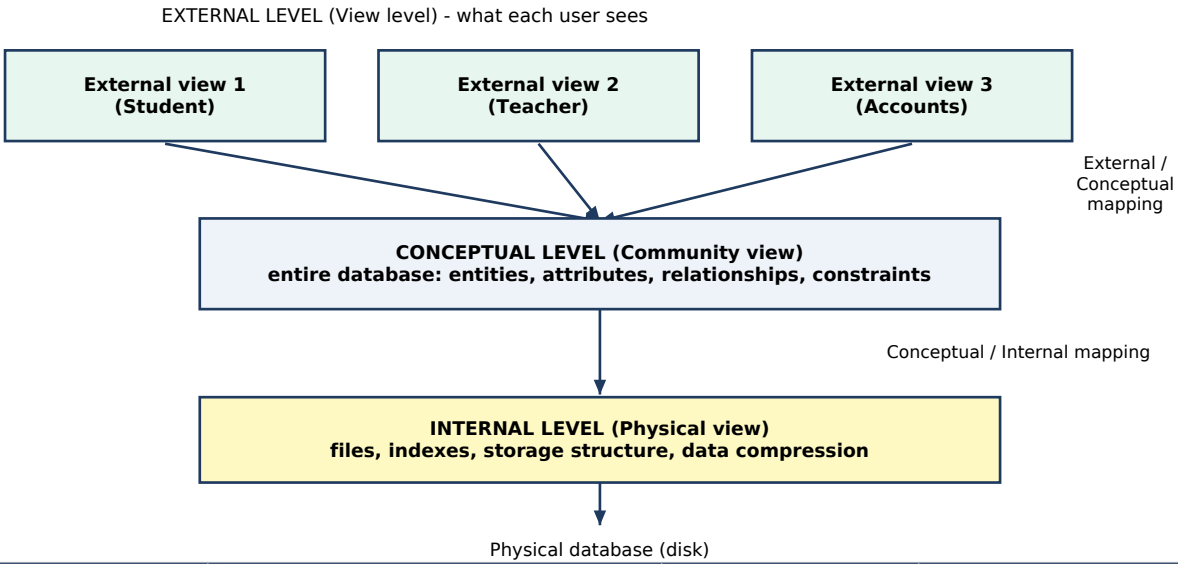
1.4 Schema, sub-schema, instances

- **Schema:** the complete logical structure of the whole database. Also called **conceptual schema**.
- **Sub-schema:** a **part / subset of the schema** that is meant for one user or application (also called external schema / user view). A sub-schema shows only the fields the user needs. *Example: the accounts section sees only STUDENT(Roll_no, Name, Fees), not marks.*
- **Instance:** the data in the database at a given moment (a snapshot); changes with every update. Schema stays the same while instances change.

1.5 Data abstraction and three-level ANSI-SPARC architecture

★ BOTH papers: W2025 Q1(c)OR [7M: data abstraction + three-level ANSI-SPARC architecture] + S2026 Q1(b) [4M: data abstraction + 3 levels]

Data abstraction means **hiding the complex details of how data is stored** and showing the user only what is needed. It makes the system easy to use. The **ANSI-SPARC three-level architecture** gives three levels of abstraction:



Level	Describes	Who uses it	Example
1. Internal (physical) level	**HOW** the data is physically stored on disk: file structure, indexes, record layout, compression, access paths	DBMS and system programmers / DBA	Student file is stored as B-tree index on Roll_no, record of 100 bytes
2. Conceptual (logical) level	**WHAT** data is stored and the relationships among them: all entities, attributes, constraints for the whole database. No storage detail.	DBA and database designers	STUDENT(Roll_no, Name, City, Branch, Fees)
3. External (view) level	The part of the database a **particular user group** sees; many different views possible; hides rest of data	End users, application programmers	Teacher sees (Roll_no, Name, Marks); Accountant sees (Roll_no, Fees)

Mappings: the DBMS converts requests between levels - *external/conceptual mapping* and *conceptual/internal mapping*. This is what gives **data independence**.

Exam tip: 4 marks: write the definition + three levels with one line each. 7 marks: add diagram, table, mappings and data independence.

1.6 Data independence

Data independence is the ability to **change the schema at one level without having to change the schema at the next higher level**.

Type	Meaning	Example
Physical data independence	Change the internal (physical) schema without changing the conceptual schema or application programs	Moving the database to a new disk, adding an index, changing file organisation
Logical data independence	Change the conceptual schema without changing external views or programs (harder to achieve)	Adding a new column or table to the database - old user views still work

Last-minute revision sheet - Unit 1

- Data = raw facts; information = processed data; metadata = data about data; schema = design; instance = data at a moment.
- DBMS advantages: less redundancy, consistency, sharing, security, integrity, concurrency, backup, data independence.
- DBMS disadvantages: cost, complexity, trained staff, size, performance overhead, single point of failure.
- DBA: schema definition, storage structure, authorisation, security, backup / recovery, tuning, maintenance, user liaison.
- ANSI-SPARC levels: Internal (how stored) - Conceptual (what stored) - External (user views).
- Data independence: physical (change storage) and logical (change conceptual schema) without changing upper level.